Tomoka State Park Living Shoreline Stabilization

Volusia County, Florida

FY 18/19

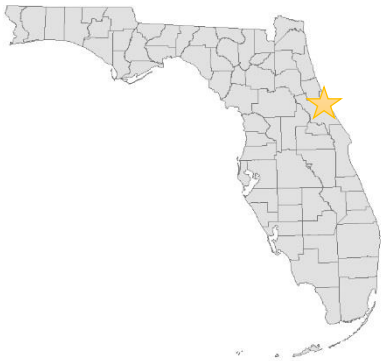
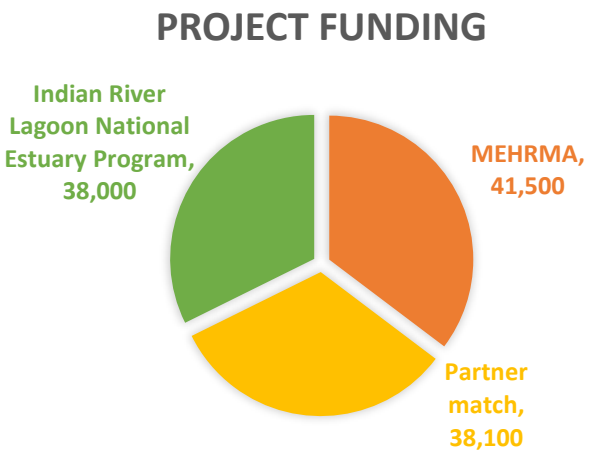


Figure 1. Location of Tomoka State Park



Project Description

Tomoka State Park spans 2,000 acres in northern Volusia County, and is home to important historical, ecological, and cultural resources (Figure 1). The Park is located along a unique large estuarine basin, and provides opportunities for camping, fishing, boating, kayaking, wildlife viewing, and hiking. Numerous Native American shell middens and plantation-era artifacts are protected within the Park, however, segments of shoreline are severely eroding due to several strong storms and constant wave action from boat wakes. This project stabilized 1000 meters of shoreline along Tomoka State Park using 'Living Shoreline' principles. The Living Shoreline design consisted of mangroves (*Rhizophora mangle, Avicennia germinans*, and *Laguncularia racemosa*), smooth cordgrass (*Spartina alterniflora*), and oyster shell bags. Living shoreline components were selected due to their natural occurrence in the region and were monitored for one year following installation. Vegetated shorelines and oyster reefs provide numerous benefits, including water filtration, nutrient cycling, and carbon sequestration. A combination of shoreline plants and oyster reefs trap sediment, allowing shorelines to stabilize and maintain their position. Living shorelines can naturally adjust to impacts of climate change such as sea level rise and increased storm activity, and preserve essential ecological functions in the future. Local volunteers and students were heavily engaged in installing and monitoring the project.

Benefits

- **Stabilize eroding shoreline along sensitive cultural resources.**
- **Provide 0.75 acres of intertidal habitat for fish and wildlife, including mangroves, salt marsh grasses, and oyster reef.**
- **Increase water filtration, sediment trapping, and coastal resilience using living shoreline techniques.**
- **Engage the community in habitat conservation.**

FWC Strategic Plan Elements Supported

Theme 1, Goal 2. Ensure that sufficient habitat exists to support healthy populations of all native species of fish and wildlife.

Strategy:

- Manage habitats to sustain healthy and diverse fish and wildlife populations.

Theme 2, Goal 1. Provide residents and visitors with quality fishing, hunting, boating, and wildlife viewing opportunities that meet their needs and expectations while providing for the sustainability of the natural resources involved.

Strategy:

- Develop, acquire, and use appropriate biological and sociological science to provide fishing, hunting, boating, and wildlife-viewing opportunities that meet the needs and expectations of user groups while ensuring sustainability of fish and wildlife resources.

Theme 3, Goal 3. Increase opportunities for residents and visitors, especially youth, to actively support and practice fish and wildlife conservation stewardship.

Strategies:

- Inform residents and visitors of conservation stewardship and encourage them to become actively involved in the conservation of fish and wildlife.
- Provide and promote opportunities for residents and visitors, especially youth, to participate in conservation stewardship activities, including volunteer opportunities with the FWC.

Acreage restored: 0.75 acres of intertidal habitat.



Figure 2: Volunteers assist with installation of living shoreline at Tomoka State Park..



Figure 3: Oyster shell bags help reduce wave energy, protect vegetation and trap sediment along the shoreline.